

Balancing Redox Equations 3

Essential Pre-Uni Chemistry K3.3

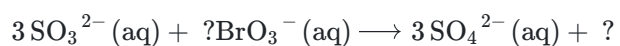
Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level P2

Complete the balanced equations to show the reactions between the following pairs of substances in alkaline aqueous conditions (no fractions).

Part A

Sulfite and bromate(V)

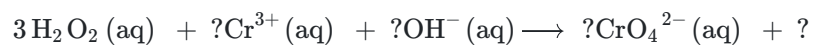
sulfite and bromate(V)



Part B

Hydrogen peroxide and chromium(III)

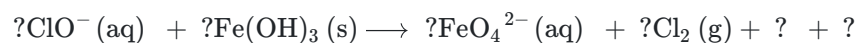
hydrogen peroxide and chromium(III)



Part C

Chlorate(I) and iron(III)

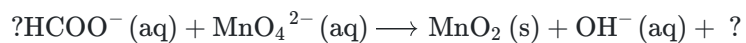
chlorate(I) and iron(III)



Part D

Manganate(VI) and methanoate

manganate(VI) and methanoate



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Disproportionation 2

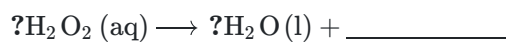
Essential Pre-Uni Chemistry K4.2

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level P1

Complete and balance the following equations that represent disproportionation reactions.

Part A

(a)



Part B

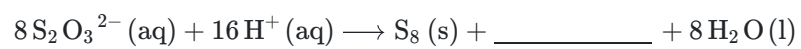
(b)



Please **click on and drag** the pre-loaded species in the equation editor to create your chemical equation.

Part C

(c)



Please **click on and drag** the pre-loaded species in the equation editor to create your chemical equation.

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Sodium Nitrite

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level C1

Nitrogen can be obtained in the laboratory by warming a mixture of ammonium chloride and sodium nitrite, NaNO_2 . Water is also produced and a solid is left.

Part A

Ammonium chloride and sodium nitrite

Suggest an identity for the solid.

Part B

Equation

Write a balanced equation for the reaction, including state symbols, balancing to obtain the lowest integer coefficients possible.

Part C

Ammonium chloride and sodium nitrate

A similar reaction takes place when ammonium chloride is heated with sodium nitrate NaNO_3 but this time the only different product is an oxide of nitrogen. Suggest a formula for this oxide, and the oxidation state of nitrogen in it.

Formula:

Oxidation state:

Adapted with permission from UCLES, A-Level Chemistry, June 1994, Paper 1, Question 7

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Calorimetry 1

Essential Pre-Uni Chemistry F1.1

Subject & topics: Chemistry | Physical | Energetics **Stage & difficulty:** A Level C1

Specific heat capacity of water = $4.18 \text{ J g}^{-1} \text{ K}^{-1}$.

Part A

(a)

Calculate the heat capacity of an object with mass 1.80 kg and specific heat capacity $0.32 \text{ J g}^{-1} \text{ K}^{-1}$.

Part B

(b)

Calculate the heat capacity of a calorimeter if its temperature is raised 2.5 K by 35 kJ of heat.

Part C

(c)

Calculate the expected increase in temperature when 2.4 kJ of heat is transferred to a calorimeter of heat capacity 720 J K^{-1} .

Part D

(d)

Calculate the heat required to raise the temperature of a calorimeter of heat capacity 1.6 kJ K^{-1} by 3.8°C .

Part E

(e)

Calculate the specific heat capacity of a calorimeter if it has a mass of 375 g and its temperature is raised 4.2 K by 2160 J of heat.

Part F

(f)

Calculate the heat required to raise the temperature of 3.14 kg of water by 12.2 K .

Part G

(g)

Calculate the mass of water whose temperature is raised through 16.0 K by 6.7 kJ of heat.

Part H

(h)

A calorimeter consists of 140 g of aluminium and 300 g of water. 6100 J of heat raises its temperature by 4.42 K. Calculate to three significant figures:

its heat capacity;

the specific heat capacity of aluminium.

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Calorimetry 9

Essential Pre-Uni Chemistry F1.9

Subject & topics: Chemistry | Physical | Energetics **Stage & difficulty:** A Level P1

Specific heat capacity of water = $4.18 \text{ J g}^{-1} \text{ K}^{-1}$.

When 5.0 g of ammonium nitrate dissolves in 100 g of water, the temperature of the water drops from 18°C to 14°C . Calculate the enthalpy of solution of ammonium nitrate in kJ mol^{-1} using the following scheme.

Part A

Formula

Write down the formula of ammonium nitrate.

Part B

Relative formula mass

Calculate the relative formula mass of ammonium nitrate. Give your answer to 3 significant figures.

Part C

Number of moles

Calculate the number of moles of ammonium nitrate in 5.0 g.

Part D

Heat loss of water

Calculate the heat lost from the 100 g of water. Give your answer to 2 significant figures.

Part E

Molar heat loss of ammonium nitrate

Calculate the heat lost per mole of ammonium nitrate.

Part F

Enthalpy of ammonium nitrate

Give the enthalpy of solution of ammonium nitrate.

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Homologous Series: Combustion

Subject & topics: Chemistry | Physical | Energetics **Stage & difficulty:** A Level C2

A student was studying the enthalpies of combustion of different alkanes, and thinking about the pattern seen when descending the homologous series.

Part A

General equation

The student started by writing a balanced chemical equation for the combustion of a general alkane

C_nH_{2n+2} , by balancing for one mole of the alkane:



Find the correct expressions for x , y and z as a function of n .

x :

The following symbols may be useful: n

y :

The following symbols may be useful: n

z :

The following symbols may be useful: n

Part B

Bonds broken

Next, the student was thinking about all the bonds present on the left-hand side of the equation previously balanced, which need to be broken in order to form the products. How many of the different types of bond are present, per molecule of C_nH_{2n+2} , assuming a correctly balanced equation?

Number of C—C bonds:

The following symbols may be useful: n

Number of C—H bonds:

The following symbols may be useful: n

Number of O=O bonds:

Part C

Bonds formed

The following consideration concerned all the bonds present on the right-hand side of the equation previously balanced, i.e. the bonds formed during the reaction. How many of the different types of bond are present, per molecule of C_nH_{2n+2} , assuming a correctly balanced equation?

Number of C=O bonds present:

The following symbols may be useful: n

Number of O–H bonds present:

Part D

Bond enthalpy assumption

When using average bond enthalpies to estimate enthalpies of reaction, we must assume that the average values used are representative of the bonding present (we can use different average values, e.g. specific to hydrocarbons to improve estimates). What other assumption about the chemicals present must be met for a calculation using bond enthalpies alone to give a good estimate of the enthalpy of reaction?

Part E**Bond enthalpy calculation**

Using the following bond enthalpies, find an expression for the enthalpy change (in kJ mol^{-1}) of the reaction previously written out (corresponding to the complete combustion of $\text{C}_n\text{H}_{2n+2}$) assuming it is carried out under conditions so that the assumption from the previous part is met. Do not include the units in the expression, and quote numerical values in the expression to 3 s.f.

Bond	Bond enthalpy / kJ mol^{-1}
C–C	348
C–H	412
O=O	498
C=O in CO_2	805
O–H	463

The following symbols may be useful: n

Part F**Deviation at RTP**

If the combustion is instead carried out at RTP (room temperature and pressure), how would you expect the empirically obtained enthalpy change of combustion of ethane to compare to that calculated using the formula derived above?

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Electrical Compensation Calorimeter

Subject & topics: Chemistry | Physical | Energetics **Stage & difficulty:** A Level C1

An electrical compensation calorimeter, such as that shown below, can be used to measure enthalpy changes of reactions.

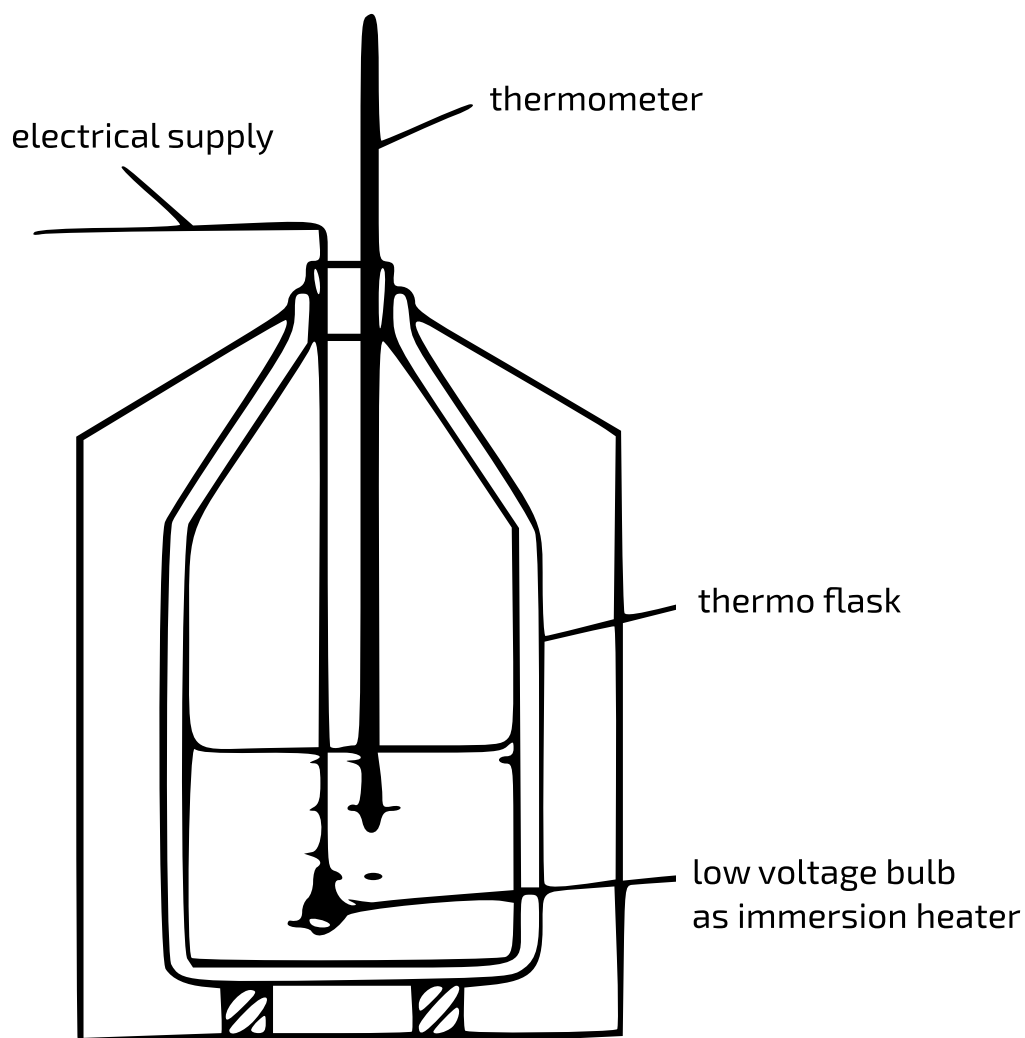


Figure 1: Electrical compensation calorimeter

Part A

Neutralisation reaction

A student used this calorimeter to measure the enthalpy change of neutralisation of hydrochloric acid by aqueous potassium hydroxide.

Write an ionic equation, including state symbols, for this reaction.

Part B

Moles

The student measured 100 cm^3 of 0.40 mol dm^{-3} HCl into the flask and then added 70 cm^3 of 0.60 mol dm^{-3} KOH and recorded a temperature rise of 3.3°C . Calculate the amount moles of each and hence conclude which reactant was in excess.

Amount in moles of hydrogen ions:

Amount in moles of hydroxide ions:

The is in excess since there is a amount in moles of hydrogen ions than hydroxide ions present.

Items:

HCl

KOH

greater

smaller

Part C**Bulb**

The student washed out the flask, placed in it 170 cm^3 of water, and supplied electrical energy to the bulb until the temperature rise was the same as before. This took 120 seconds.

The bulb supplied 20 Joules every second.

Calculate how much heat was given out by the bulb.

Part D**Enthalpy change**

The heat supplied by the bulb is equal to the heat released during the neutralisation reaction. Calculate the enthalpy change of neutralisation of hydrochloric acid by aqueous potassium hydroxide.

Part E

Other acids

The student carried out two similar experiments using different acids in place of hydrochloric acid.

When nitric acid was used, the enthalpy change of neutralisation was found to be the **same** as that obtained using hydrochloric acid.

When ethanoic acid was used the enthalpy change of neutralisation was found to be **less** exothermic than that obtained using hydrochloric acid.

Suggest why.

Hydrochloric and nitric acid are both acids, whereas ethanoic acid is a acid. Therefore O–H bonds need to be broken when hydrochloric or nitric acid is neutralised and O–H bonds are broken when ethanoic acid is neutralised as it . Bond breaking is , explaining why the enthalpy change of neutralisation is less exothermic for ethanoic acid.

Items:

Part F

Calorimeters

Suggest one advantage of using an electrical compensation calorimeter in this experiment rather than a conventional calorimeter.

- ☐ the volume of the solution is known more accurately
- ☐ the energy supplied is known more accurately
- ☐ the temperature change is known more accurately
- ☐ the starting concentrations are known more accurately

C₃H₆ Combustion

Subject & topics: Chemistry | Physical | Energetics **Stage & difficulty:** A Level P2

A and **B** are two isomers with the molecular formula C₃H₆. The standard enthalpies of formation, $\Delta_f H^\ominus$, of both **A** and **B** have been found by first measuring the standard enthalpies of combustion, $\Delta_c H^\ominus$, of each. These values are given in the table below, together with the standard enthalpies of combustion of carbon and hydrogen.

	A	B	carbon	hydrogen
$\Delta_c H^\ominus / \text{kJ mol}^{-1}$	−2058	−2091	−393.5	−241.8

Part A

Combustion equation

Give the equation for the complete combustion of C₃H₆. (Balance it for one mole of the hydrocarbon.)

Part B

$\Delta_f H^\ominus$ of A

Calculate the standard enthalpy of formation of **A**.

Part C

$\Delta_f H^\ominus$ of **B**

Calculate the standard enthalpy of formation of **B**.

Part D

Isomerisation

Gaseous **B** needs to be stored carefully since it can convert explosively to the elements, to isomer **A**, or to other hydrocarbons. Calculate the standard enthalpy change for the reaction **B** $\longrightarrow$ **A**.

Adapted with permission from the Cambridge Chemistry Challenge 2011, Question 1

Question deck:

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Ethene Combustion

Subject & topics: Chemistry | Physical | Energetics **Stage & difficulty:** A Level C1

The standard enthalpy change of combustion of but-1-ene, $\text{C}_4\text{H}_8(\text{g})$, is $x \text{ kJ mol}^{-1}$.

The standard enthalpy change of the reaction $2 \text{C}_2\text{H}_4(\text{g}) \longrightarrow \text{C}_4\text{H}_8(\text{g})$ is $y \text{ kJ mol}^{-1}$.

Write down an expression, in terms of x and y , for the standard enthalpy change of combustion of ethene, $\text{C}_2\text{H}_4(\text{g})$ when expressed in kJ mol^{-1} (your answer should not feature any units).

The following symbols may be useful: x , y

Adapted with permission from UCLES, A Level Chemistry, November 1993, Paper 4, Question 8

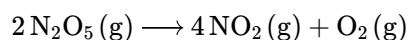
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Dinitrogen Pentoxide Formation

Subject & topics: Chemistry | Physical | Energetics **Stage & difficulty:** A Level C2

In the gas phase, N_2O_5 decomposes to oxygen and nitrogen dioxide. Use the data below (determined at 298 K) to calculate the standard enthalpy change at 298 K for the reaction:



	value/ kJ mol^{-1}
$\Delta_f H^\circ$ of $\text{N}_2\text{O}_5(\text{g})$	11.3
$\Delta_r H^\circ$ for $\text{NO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \longrightarrow \text{NO}_2(\text{g})$	-58.1
bond strength in $\text{N}_2(\text{g})$	945
bond strength in $\text{O}_2(\text{g})$	498
bond strength in $\text{NO}(\text{g})$	631

Adapted with permission from the Cambridge Chemistry Challenge, Question 1, 2019

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